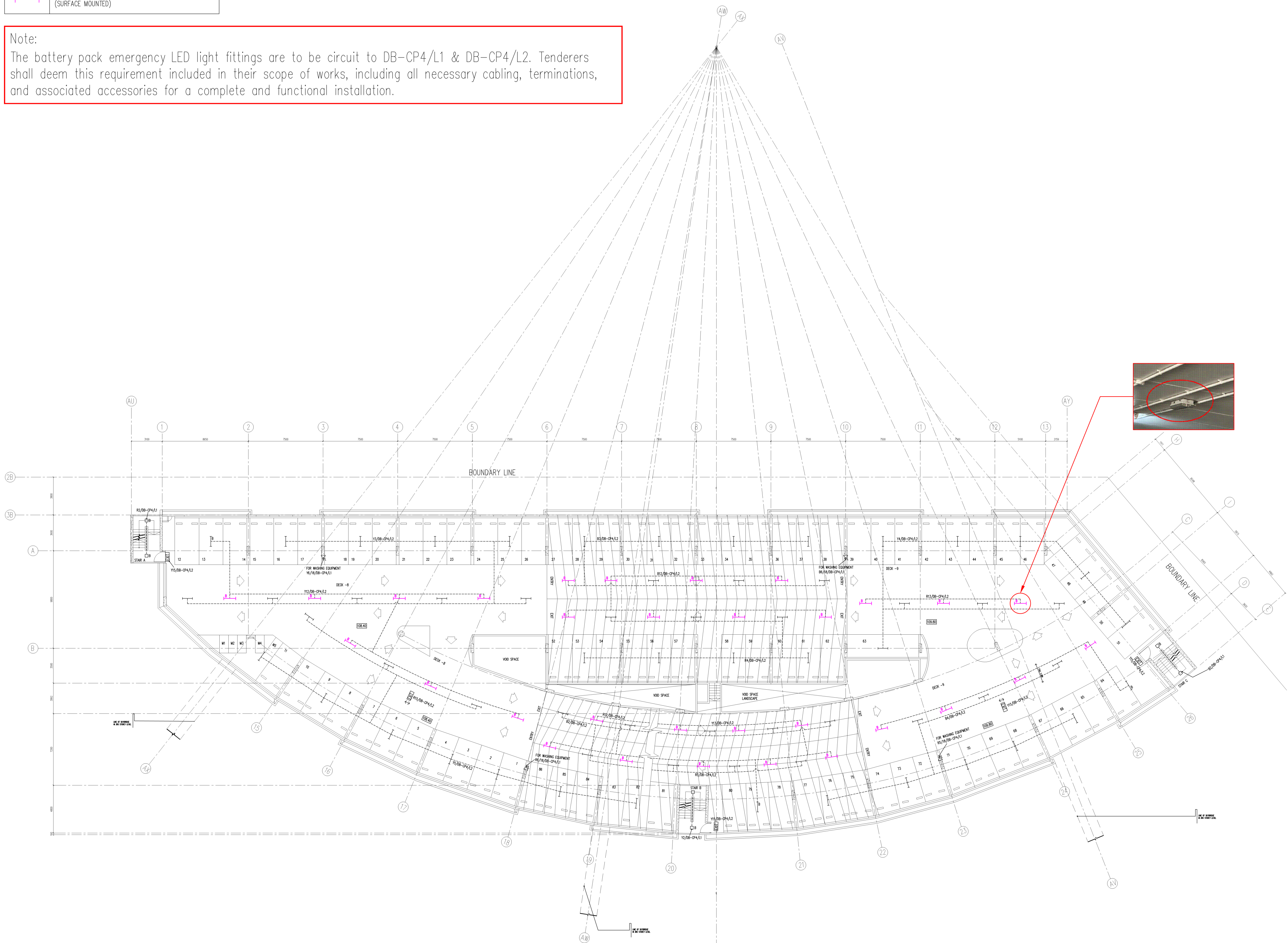


LEGEND

 LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)

Note:

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.



4TH STOREY PLAN
DECK 8 & DECK 9
SCALE 1 : 200

TENDER DRAWING
NOVEMBER 2025

REV	DATE	DRAWN	CHECKED	DESCRIPTION
-----	------	-------	---------	-------------

TENDER DRAWING

CLIENT:



HDB HUB 480 LORONG 6 TOA PAYOH SINGAPORE 310480

CLIENT SIGNATURE:

DRAWING NO. HB-N2303-B79-E-205 REVISION: -

PROJECT TITLE:

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)

ELECTRICAL INSTALLATION
LIGHTING AND POWER LAYOUT
4TH STOREY PLAN DECK 8 & DECK 9

SCALE: AS SHOWN DATE: NOVEMBER 2025

DRAWN BY: TLY CHECKED BY: JL/NC

M & E CONSULTANT:



CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheetoon.com

C&S ENGINEER:



(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsultants.com

QUANTITY SURVEYOR:

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

DRAWING NO. HB-N2303-B79-E-205 REVISION: -

DESIGN ENGINEER

QUALIFIED PERSON

LEGEND	
	LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.

[illegible]

CLIENT:

UDD, UUR, 100, LORONG 6, TOA PAYOH, SINGAPORE 310180

DRAWING NO. HR-N2303-B79-F-206		REVISION: -
--------------------------------	--	-------------

MECHANICAL AND ELECTRICAL
CYCLOICAL IMPROVEMENT WORKS

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)

SCALE:	AS SHOWN	DATE:	NOVEMBER 2025
--------	----------	-------	---------------

M & E CONSULTANT:	
-------------------	--

CCA & Partners Pte. Ltd.

C&S ENGINEER:

 **ANG & ONG**
CONSULTANTS PTE LTD

Blk 28, Sin Ming Lane,
#07-137, Midview City,

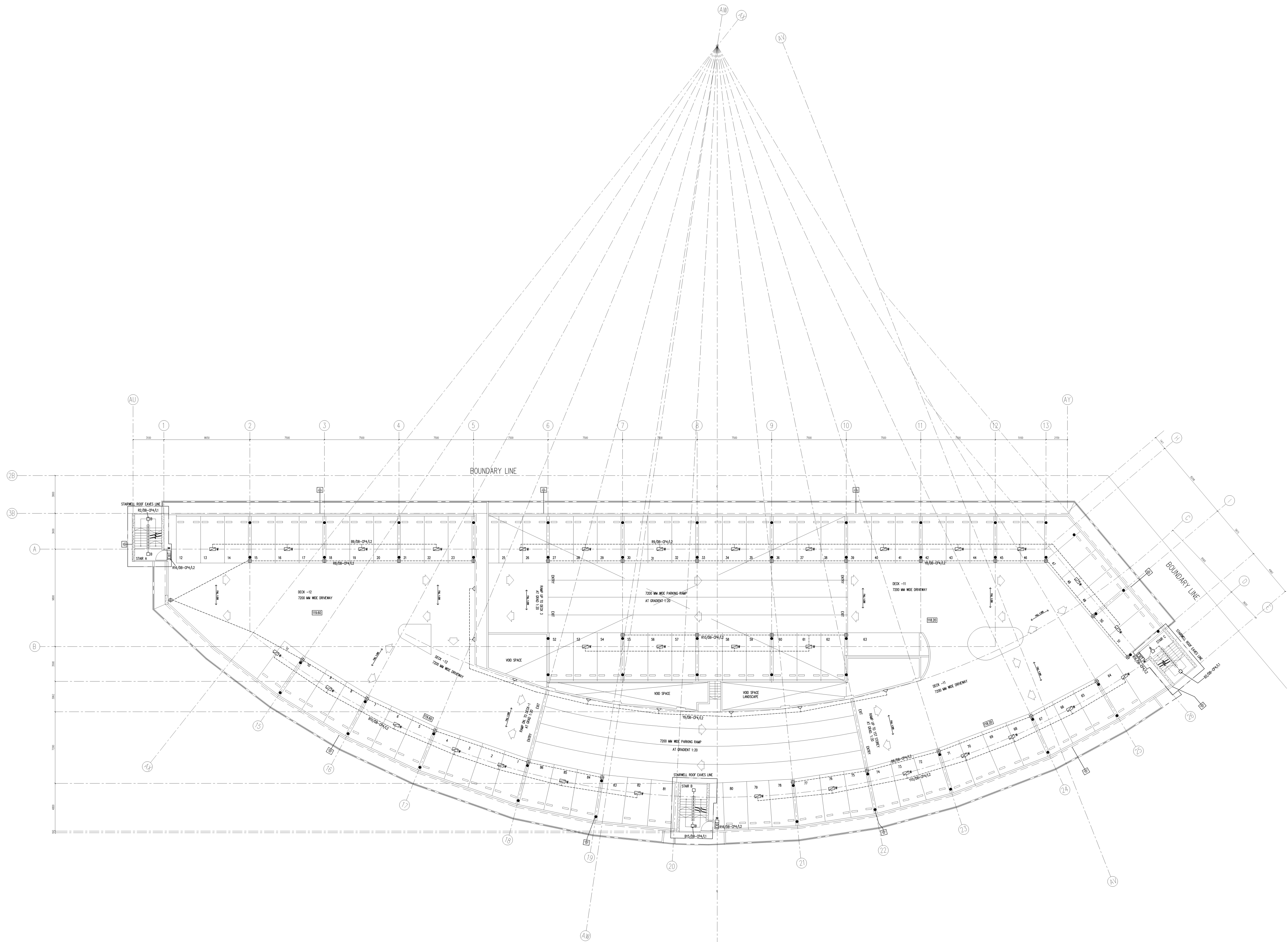
QUANTITY SURVEYOR:

OS Consultants Pty Ltd

21 Bukit Batok Crescent
#16-70 WCEGA Tower

DRAWING NO. HB-N2303-B79-E-206	REVISION: -

DESIGN ENGINEER	QUALIFIED PERSON
-----------------	------------------



ROOF STOREY PLAN
DECK 11 & DECK 12
SCALE 1 : 200

TENDER DRAWING
NOVEMBER 2025

REV	DATE	DRAWN	CHECKED	DESCRIPTION
-----	------	-------	---------	-------------

TENDER DRAWING

CLIENT:



HOUSING &
DEVELOPMENT
BOARD

HDB HUB 480 LORONG 6 TOA PAYOH SINGAPORE 310480

CLIENT SIGNATURE:

DRAWING NO. HB-N2303-B79-E-207

REVISION: -

PROJECT TITLE:

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772MSCP)

ELECTRICAL INSTALLATION
LIGHTING, POWER AND LIGHTNING PROTECTION LAYOUT
ROOF STOREY PLAN DECK 11 & DECK 12

SCALE:

AS SHOWN

DATE:

NOVEMBER 2025

DRAWN BY:

TLY

CHECKED BY:

JL/NC

M & E CONSULTANT:

CCAPpartners

CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheetoon.com

C&S ENGINEER:

ANG & ONG
CONSULTANTS PTE LTD

(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsultants.com

QUANTITY SURVEYOR:

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

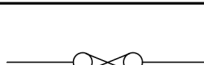
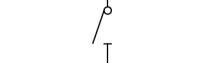
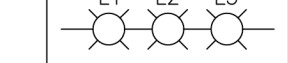
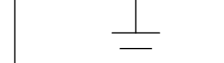
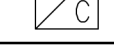
DRAWING NO. HB-N2303-B79-E-207

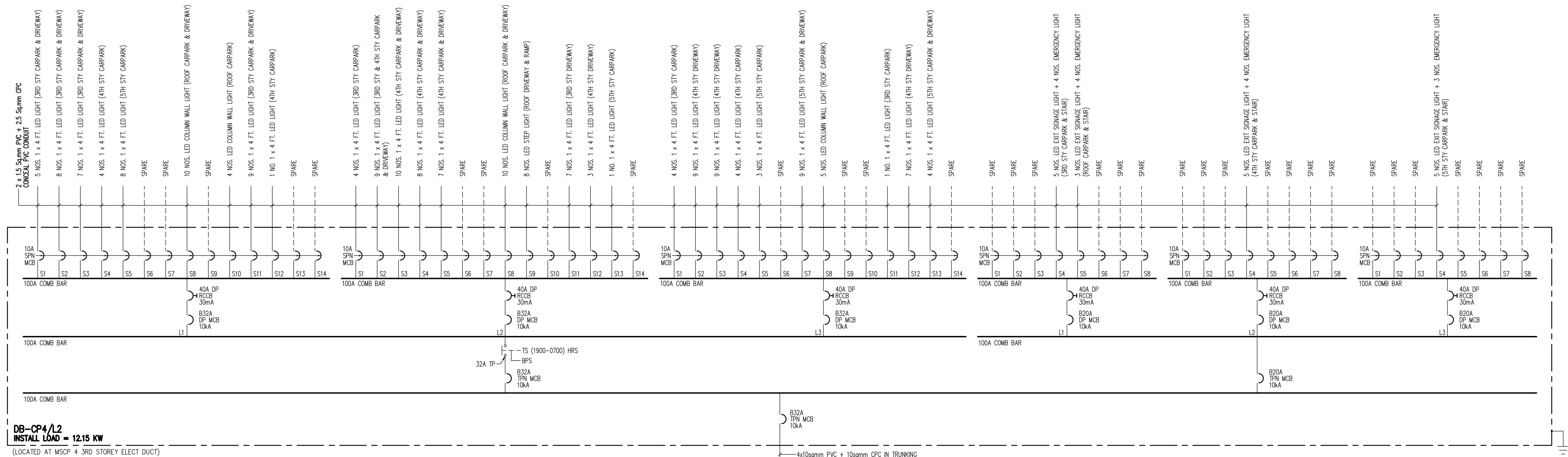
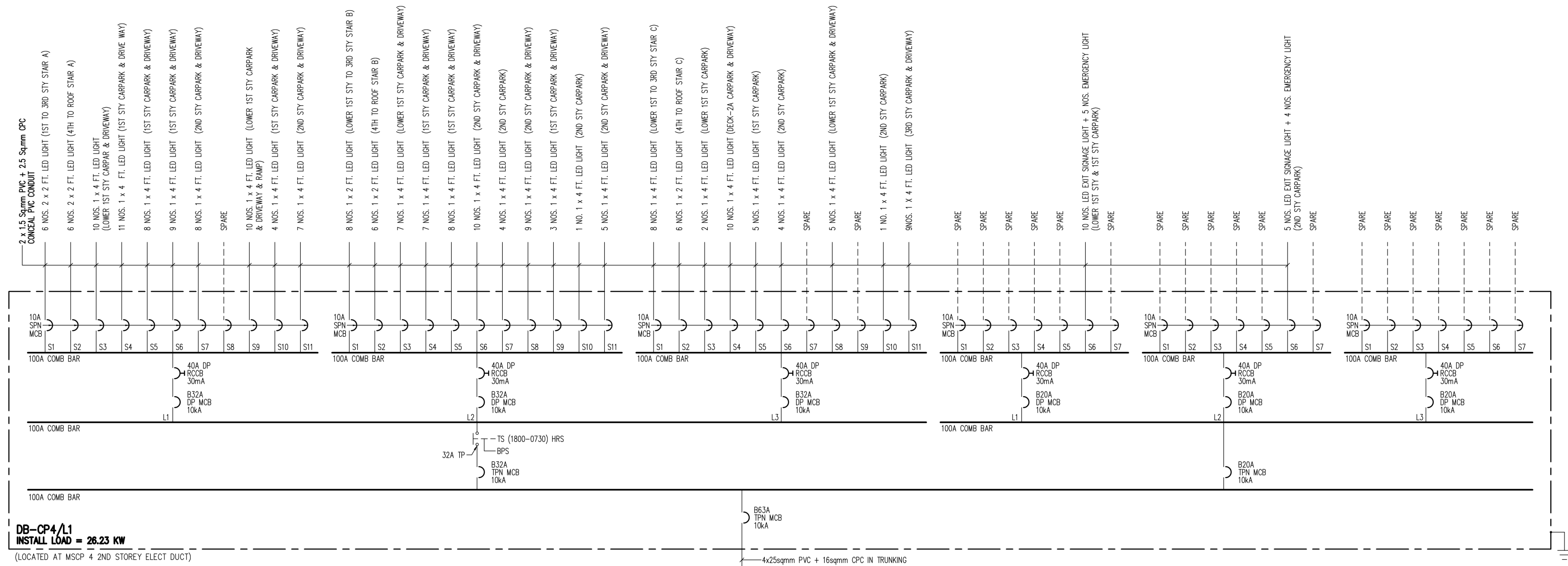
REVISION: -

DESIGN ENGINEER

QUALIFIED PERSON

LEGEND

	HRC FUSE (RATING AS SHOWN)		AMMETER		INVERSE DEFINITE MINIMUM TIME-LAG EARTH FAULT RELAY OF 3/10 CHARACTERISTIC C/W INSTANTANEOUS HIGH SET ELEMENT
	MOULDED CASE CIRCUIT BREAKER/ MINIATURE CIRCUIT BREAKER		VOLTMETER		OVERCURRENT RELAY
	RESIDUAL CURRENT CIRCUIT BREAKER		VOLTMETER SELECTOR SWITCH (L1L2-L2L3-L3L1, L1N, L2N, L3N, OFF)		EARTH FAULT RELAY
	ISOLATOR		AMMETER SELECTOR SWITCH (L1-L2-L3-OFF)		INDICATING LIGHT (LED)
	CONTACTOR		CURRENT TRANSFORMER		EARTHING
	KILOWATT-HOUR METER		ZERO PHASE SEQUENCE CURRENT TRANSFORMER	TS	TIME SWITCH
	DIGITAL POWER METER		EARTH LEAKAGE RELAY C/W TIME LAG (10A TYPE)	BPS	BY-PASS SWITCH
	AMMETER C/W MAXIMUM DEMAND INDICATOR		INVERSE DEFINITE MINIMUM TIME-LAG OVERCURRENT RELAY OF 3/10 CHARACTERISTIC C/W INSTANTANEOUS HIGH SET ELEMENT		



TENDER DRAWING
NOVEMBER 2025

REV	DATE	DRAWN	CHECKED	DESCRIPTION
-----	------	-------	---------	-------------

TENDER DRAWING

CLIENT:



**HOUSING &
DEVELOPMENT
BOARD**

HDB HUB 480 LORONG 6 TOA PAYOH SINGAPORE 310480

CLIENT SIGNATURE:

DRAWING NO. HB-N2303-B79-E-302	REVISION: -
--------------------------------	-------------

PROJECT TITLE:

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)

ELECTRICAL INSTALLATION

LV SINGLE LINE DIAGRAM 2

SCALE: N T S	DATE: NOVEMBER 2025
DRAWN BY: TLY	CHECKED BY: JL/NC

M & E CONSULTANT:



CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheechoon.com

C&S ENGINEER:



(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsults.com

QUANTITY SURVEYOR

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

DRAWING NO. HB-N2303-B79-E-302	REVISION: -
--------------------------------	-------------

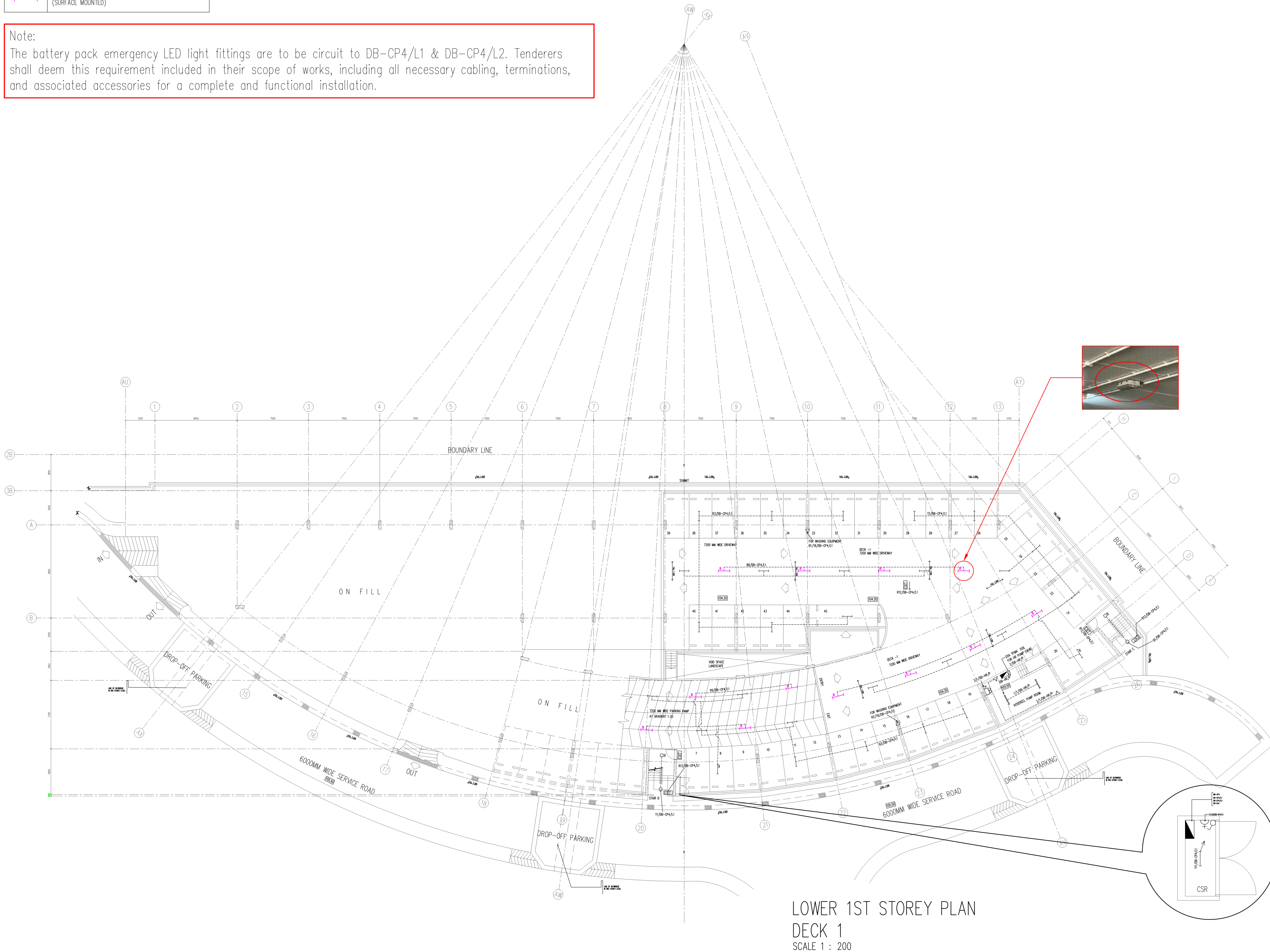
DESIGN ENGINEER	QUALIFIED PERSON
-----------------	------------------

LEGEND

 LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)

Note:

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.



LOWER 1ST STOREY PLAN
DECK 1
SCALE 1 : 200

TENDER DRAWING
NOVEMBER 2025

REV	DATE	DRAWN	CHECKED	DESCRIPTION

TENDER DRAWING

CLIENT:



HDB HUB 480 LORONG 6 TOA PAYOH SINGAPORE 310480

CLIENT SIGNATURE:

DRAWING NO. HB-N2303-B79-E-201 REVISION: -
PROJECT TITLE:

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)
ELECTRICAL INSTALLATION
LIGHTING AND POWER LAYOUT
LOWER 1ST STOREY PLAN DECK 1

SCALE: AS SHOWN DATE: NOVEMBER 2025
DRAWN BY: TLY CHECKED BY: JL/NC

M & E CONSULTANT:



CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheetoon.com

C&S ENGINEER:



(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aaoconsultants.com

QUANTITY SURVEYOR:

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

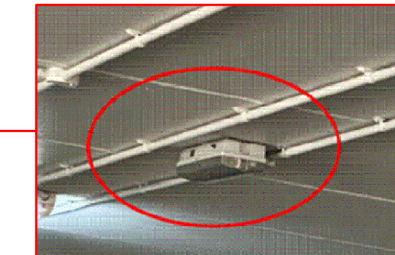
DRAWING NO. HB-N2303-B79-E-201 REVISION: -

DESIGN ENGINEER

QUALIFIED PERSON

	LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)
---	--

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.

[illegible]

CLIENT:

 HOUSING &
DEVELOPMENT
BOARD

CLIENT SIGNATURE: _____

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

SCALE:	AS SHOWN	DATE:	NOVEMBER 2025
DRAWN BY:	TLY	CHECKED BY:	JL/NC

CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheechoon.com

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsults.com

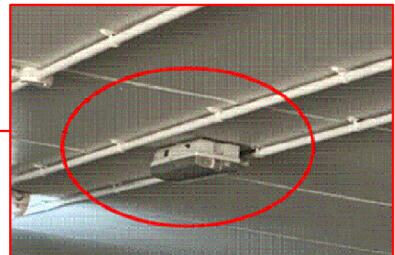
QS Consultants Pte Ltd

DRAWING NO. HB-N2303-B79-E-202	REVISION: -
--------------------------------	-------------

DESIGN ENGINEER	QUALIFIED PERSON
-----------------	------------------

LEGEND	
	LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.



2ND STOREY PLAN
DECK 4 & DECK 5
SCALE 1 : 200

TENDER DRAWING
NOVEMBER 2025

TENDER DRAWING

CLIENT:

HOUSING &
DEVELOPMENT
BOARD

HDB HUB 480 LORONG 6 TOA PAYOH SINGAPORE 310480

CLIENT SIGNATURE:

DRAWING NO. HB-N2303-B79-E-203

PROJECT TITLE:

MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)
ELECTRICAL INSTALLATION
 LIGHTING AND POWER LAYOUT
 2ND STOREY PLAN DECK 4 & DECK 5

SCALE:	AS SHOWN	DATE:	NOVEMBER 2025
DRAWN BY:	TLY	CHECKED BY:	JL/NC

M & E CONSULTANT:



CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheechoon.com

C&S ENGINEER:



(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsults.com

QUANTITY SURVEYOR:

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

DRAWING NO. HB-N2303-B79-E-203

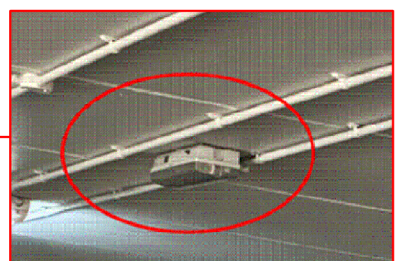
--	--

DESIGN ENGINEER

QUALIFIED PERSON

LEGEND	
	LED BULKHEAD LIGHT FITTING C/W BATTERY PACK (SURFACE MOUNTED)

The battery pack emergency LED light fittings are to be circuit to DB-CP4/L1 & DB-CP4/L2. Tenderers shall deem this requirement included in their scope of works, including all necessary cabling, terminations, and associated accessories for a complete and functional installation.



TENDER DRAWING
NOVEMBER 2025

REV	DATE	DRAWN	CHECKED	DESCRIPTION
TENDER DRAWING				

TENDER DRAWING

 HOUSING &
DEVELOPMENT
BOARD

CLIENT SIGNATURE

PROJECT TITLE:

PROJECT TITLE:
MECHANICAL AND ELECTRICAL
CYCLICAL IMPROVEMENT WORKS
TO MULTI-STOREY CAR PARKS
(BATCH 34)

DRAWING TITLE: (FOR EST BLK NO 772AMSCP)
ELECTRICAL INSTALLATION
LIGHTING AND POWER LAYOUT
3RD STOREY PLAN DECK 6 & DECK 7

SCALE:	AS SHOWN	DATE:	NOVEMBER 2025
DRAWN BY:	TLY	CHECKED BY:	JL/NC

CCAPartners

CCA & Partners Pte Ltd
M&E Consulting Engineer
46 Lorong 17 Geylang, #02-02, Enterprise Industrial Building,
Singapore 388568, Tel: 6476 2272 Fax: 6475 6229
E-mail : cca@cheechoon.com

C&S ENGINEER:

ANG & ONG
CONSULTANTS PTE LTD

(Professional Engineers, Project Management and
and Structural Inspection)

Blk 28, Sin Ming Lane,
#07-137, Midview City,
Singapore 573972
Tel: (65) 6459 5438 Fax: (65) 6459 5267
Email: info@aoconsults.com

QS Consultants Pte Ltd

21 Bukit Batok Crescent
#16-70, WCEGA Tower
Singapore 658065
Tel: 6899 4298 Fax: 6899 4981
E-mail : qs@qsconsultants.sg

DRAWING NO. HB-N2303-B79-E-204	REVISION: -
--------------------------------	-------------

QUALIFIED PERSON
